Akshay Patil

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EDUCATION

Georgia Institute of Technology – Atlanta, GA **August 2026 – May 2029**
Bachelor of Science in Aerospace Engineering | Sophomore | GPA: 4.00
Current coursework: CS 1371 MATLAB, MATH 2551 Multivariable Calculus, COE 2001 Statics, ME 1670 SolidWorks, AE 1601 Intro to Aerospace Engineering, MSE 2001 Materials

School of Science and Engineering at Townview – Dallas, TX **August 2022 – May 2026**
3.97 Unweighted GPA | Rank 2 | 1600 SAT | National Merit Scholar | NASA High School Aerospace Scholar

EXPERIENCE

Hardpoint Robotics | Co-Founder **June 2026 – Present**
- Hardpoint builds NDAA-compliant development UAVs, aircraft for customers who are not allowed to fly foreign-sourced hardware
- Designing airframes in CAD and 3D printing CF-Nylon parts for them
- Working with various North American vendors to manufacture the carbon fiber structures and source compliant components, where lead times shape the design as much as the loads do

Salvo Dynamics | Design Engineer **August 2026 – Present**
- Salvo builds small interceptor aircraft meant to fly out and bring down hostile drones, and is separately building an RC jet to break the speed of sound
- Turning optimized aerobody models into load-bearing structures in CAD, and laying out the avionics that go inside them
- Designing 3D printed airframes for testing, plus the assembly jigs the production aircraft are built on

SWYFT Robotics | Engineering Design Intern **May 2025 – May 2026**
- SWYFT sells competition drivetrain and structural parts to FIRST Tech Challenge and FIRST Robotics Competition teams
- Designed sprockets, gears, and pulleys that hold up under competition loads at a price teams can afford

PROJECTS

Autonomous House-Mapping Robot | Personal Project **2025 – Present**
- Jetson Orin Nano robot that explores a house with no operator and counts windows, using an MPPI controller, RTAB-Map visual SLAM, and frontier exploration I wrote
- No lidar: obstacles come from monocular depth classified against a fitted ground plane, windows from three open-vocabulary detectors filtered by a vision-language model

Pickleball Ball Boy UAV | Personal Project **May 2025 – Present**
- Benchmarked five ball detection and four localization methods; AprilTag localization was the most accurate by a wide margin
- Split the drag model into horizontal and vertical coefficients, bringing landing prediction inside a 20 cm zone

RESEARCH AND CLUBS

Hypersonics VIP, Georgia Tech | Data and Communications Team **August 2026 – Present**
- Building a sonic fatigue test facility, where acoustic loading is used to find how structures fail under sustained high-intensity sound
- Designing the sensor package that measures pressure, temperature, and airflow inside the facility

FIRST Tech Challenge Team 7172 | Captain and Lead Designer **September 2021 – May 2026**
- Lead designer on four robots; part of the Finalist Alliance at the 2026 FIRST World Championship
- Designed mechanisms using linkages, counter-springing, pulley systems, and geartrains
- Used industrial arm kinematics to design and build a 6-DoF arm robot that expands from an 18 in cube to 13 ft of reach
- 24 competition wins (2021 to 2026), plus Inspire, Think, Connect, Innovate, Design, Motivate, and Control awards

SKILLS

Robotics: ROS 2 (Nav2), RTAB-Map visual SLAM, MPPI control, AprilTag localization
Programming: Python (NumPy, Pandas, TensorFlow), MATLAB, JavaScript, SQL, Git
Hardware: NVIDIA Jetson, Raspberry Pi, Arduino, REV Control Hub
CAD: SolidWorks, Fusion 360, Onshape, Autodesk Inventor, 3D printing
Certifications: FAA Part 107 Remote Pilot, Autodesk Certified User (Inventor)